

# IAS2026 / TX4 Research Dossier

Theory, equations, thesis, claims, experiments, results, and open items  
of the synchronous-to-inverter replacement-portfolio programme

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## Abstract

This dossier is a self-contained restatement of the research programme for a collaborator (human or language model) who has lost the conversational context. It collects, in one place: the research question and thesis; the frozen model and its equations; every mathematical statement with its proof status; the experiment catalogue; the key numerical results with their source files; the canonical claim ledger with explicit status labels; the preserved negative results; the independent-tool validation status; and what is still missing before the TPWRS submission. Every number below is copied from a frozen result file or a committed document; nothing new was computed for this dossier.

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## 1 How to use this dossier

**Canonical sources.** When this dossier and a file disagree, the file wins.

- Claim ledger: docs/20260911\_CANONICAL\_THEORY\_LEDGER.md, results/20260911\_CLAIM\_MATRIX.csv (44 claims).
- Evidence: docs/FINAL\_TRANSACTION\_THEORY\_AND\_EVIDENCE.md, results/FINAL\_EVIDENCE\_TABLE.csv, docs/FINAL\_TPWS\_CLAIMS.md.
- Cross-tool validation: docs/FINAL\_CROSS\_TOOL\_VALIDATION.md (commit 0a4e18c2).
- Connected-cumulant extension: theory/FINITE\_AMPLITUDE\_CONNECTED\_CUMULANTS.md, docs/CONNECTED\_CUMULANTS\_EXTENSION.md (commits de09db3b, cdfcefcf).
- Manuscript draft: reports/papers/tx4\_policy\_dependent\_incompatibility/main.tex (11 pp, IEEE-tran).

All paths are relative to reports/poster/ias2026/research/ unless stated.

**Working rules of the programme.**

- Frozen results are never rewritten; new work goes to new files and new commits.
- Every new campaign is preregistered (rules, thresholds, hypotheses) and committed *before* it is run.
- No push to the remote. Commits use explicit pathspecs because about 888 staged files belong to other work.
- No retuning to obtain agreement; negative results are recorded and not reinterpreted.
- Status vocabulary: PROVED, NUMERICALLY VALIDATED, BENCHMARK-SPECIFIC, CONDITIONAL, REFUTED, NOT YET TESTED. An IEEE-39 observation is never promoted to a theorem.
- Mandatory scope sentence: *“The frozen IEEE-39 model has no primary frequency restoration and is analysed in relative / transverse coordinates. Absolute common-frequency restoration is outside this benchmark.”*
- MW always means active dispatch ( $P_g$ ); MVA always means rating ( $S_n$ ). They are never merged.

## 2 Research question and thesis

**Question.** Synchronous generators (SGs) are replaced by grid-following inverters (GFLs) unit by unit. Small-signal stability is not additive: every proper subset of a replacement portfolio can be stable while the complete portfolio is not. Which minimal groups of replacements are mutually incompatible, why does the network make them so, and how does the converter control policy move that structure?

**Thesis (current, after the cumulant negative results).**

local device changes  $\rightarrow$  exact network-mediated collective closure  $\rightarrow$  policy-dependent minimal failing coalitions  $\rightarrow$  non-composable replacement portfolios  $\rightarrow$  exact diagnosis, planning and targeted design.

**Three manuscript contributions (TX4).**

- C1 Policy-dependent minimal incompatibility.** The minimal incompatibility hypergraph  $\mathcal{H}(\theta)$  and its minimum order  $\kappa(\theta)$ , defined exactly in transverse coordinates;  $\mathcal{H}$  changes only at imaginary-axis crossings; a target is safe in every implementation order iff it contains no hyperedge. On IEEE-39 the converter policy reorganizes which coalitions fail; with documented governors the four-unit witness disappears while policy dependence persists.
- C2 Network-closure anatomy.** Replacement actions are affine and bus-local in the descriptor; exact network closure turns them into principal-minor interactions of all orders; every boundary is a zero of the interaction factor. At ten frozen boundaries no *Boolean* truncation below the cardinality of the changing coalition reaches the zero.
- C3 Actionable boundary motion.** The port derivative gives exact boundary normals and supports iterative (Newton) retuning; a symmetry-deflated port extends the test to zero-frequency crossings.

**Not claimed.** Rank explains  $\kappa$  (refuted); a dominant network path explains a failure (non-executable); a simple cycle causes a failure (refuted); nonlinear composability as a contribution (negative result); connected cumulants as the mechanism (refuted on the flagship, Section 4.6); universality of order four (refuted).

### 3 Model (frozen IEEE-39 benchmark)

#### 3.1 Phasor DAE

$$\dot{x} = f(x, z; \delta, \theta), \quad 0 = g(x, z; \delta, \theta), \quad A = f_x - f_z g_z^{-1} g_x, \quad (1)$$

with  $x$  the device states,  $z \in \mathbb{R}^{2N}$  the rectangular bus voltages,  $g$  Kirchhoff's current law (bus admittance matrix, device current injections, constant-power loads), index one ( $g_z$  nonsingular).

#### 3.2 Synchronous generators (two-axis, own base)

$$\dot{\delta} = \omega_B(\omega - 1), \quad M\dot{\omega} = P_m - P_e - D(\omega - 1), \quad (2)$$

$$T'_{d0}\dot{e}'_q = e_{fd} - e'_q - (x_d - x'_d)i_d, \quad T'_{q0}\dot{e}'_d = -e'_d + (x_q - x'_q)i_q, \quad (3)$$

$P_e = v_d i_d + v_q i_q + r_a(i_d^2 + i_q^2)$ ;  $D = 0$  and constant  $P_m$  (no governor) in the frozen model. AVR (harmonized from IEEEEX1, first order):  $T_E \dot{e}_{fd} = K_A(v_{\text{ref}} + v_s - |V|) - e_{fd}$ . PSS (power-input washout plus lag):  $T_6 \dot{x}_5 = P_e - x_5$ ,  $T_4 \dot{x}_6 = T_5(P_e - x_5)/T_6 - x_6$ ,  $v_s = K_S x_6$ . Code: `src/ibr_cycles/models/ieee39_devices.py`, class `SynchronousMachine`.

#### 3.3 Custom grid-following inverter (10 states + 1 Q/V state)

States  $(\vartheta, x_{\text{pll}}, p_f, q_f, x_p, x_q, i_d, i_q, x_{id}, x_{iq}, x_v)$ , on the device rating, current injection  $\gamma(i_d + j i_q)e^{j\vartheta}$  on the system base ( $\gamma = \text{rating}/100 \text{ MVA}$ ). With  $v_d + j v_q = V e^{-j\vartheta}$ ,  $P = v_d i_d + v_q i_q$ ,  $Q = v_q i_d - v_d i_q$ :

$$\dot{\vartheta} = k_{p,\text{pll}} v_q + x_{\text{pll}}, \quad \dot{x}_{\text{pll}} = k_{i,\text{pll}} v_q, \quad (4)$$

$$\tau \dot{p}_f = P - p_f, \quad \tau \dot{q}_f = Q - q_f, \quad (5)$$

$$i_d^{\text{ref}} = k_p(p_{\text{ref}} - p_f) + x_p, \quad i_q^{\text{ref}} = -(k_q(q_{\text{cmd}} - q_f) + x_q), \quad (6)$$

$$\dot{x}_p = k_{ip}(p_{\text{ref}} - p_f), \quad \dot{x}_q = k_{iq}(q_{\text{cmd}} - q_f), \quad (7)$$

$$\dot{i}_d = \frac{\omega_B}{x_f}(k_i(i_d^{\text{ref}} - i_d) + x_{id} - r_f i_d), \quad \dot{i}_q = \frac{\omega_B}{x_f}(k_i(i_q^{\text{ref}} - i_q) + x_{iq} - r_f i_q), \quad (8)$$

$$\dot{x}_{id} = k_{ii}(i_d^{\text{ref}} - i_d), \quad \dot{x}_{iq} = k_{ii}(i_q^{\text{ref}} - i_q), \quad (9)$$

$$q_{\text{cmd}} = k_{p,v} g(v_{\text{ref}} - |V|) + x_v, \quad \dot{x}_v = k_{i,v} g(v_{\text{ref}} - |V|) - w(x_v - q_{\text{ref}}). \quad (10)$$

The current-loop form above is the code form  $\dot{e}_d = v_d + k_i(i_d^{\text{ref}} - i_d) + x_{id} - x_f i_q$ ,  $\dot{i}_d = (\omega_B/x_f)(e_d - v_d - r_f i_d + x_f i_q)$  after cancellation. The leak is  $w = 0.05 \text{ rad/s}$ ;  $g = 0$  is exact fixed- $Q$  operation plus one decoupled pole at  $-w$ . No current limits, DC link, modulation delay or ride-through logic. Code: class `GridFollowingConverter`.

#### 3.4 Portfolios, dispatch and policy

Candidates  $V = \{30, 33, 35, 37\}$  (the *core*); the census uses SGs 30–38 ( $2^9 = 512$  portfolios); bus 39 (interconnection) is never replaced. Replacing unit  $i$  installs a GFL with the same active and reactive dispatch and rating equal to the SG rating (*matched dispatch*), so the AC power flow is identical for every portfolio. The policy is  $\theta = (g, k, t, h)$ : Q/V gain  $g$ , common scale  $k$  of every SG AVR gain, scale  $t$  of every AVR time constant, heterogeneity exponent  $h$  ( $T_{E,i} \mapsto G(T_{E,i}/G)^h$ ). Reference points:  $\mathbf{P}_4 = (0.03625, 1.425, 1.5, 1)$  and  $\mathbf{P}_\infty = (1, 0.5, 1.5, 1)$ . Note:  $\mathbf{P}_4 \rightarrow \mathbf{P}_\infty$  changes *both* the converter Q/V gain and the SG excitation-gain scale.

| Item          | Value (per unit on the device rating)                                     |
|---------------|---------------------------------------------------------------------------|
| SG            | two-axis, power-form swing, $D = 0$ , constant $P_m$                      |
| AVR           | $K_A = 10.1$ , $T_E = 0.25 \text{ s}$ (harmonized IEEEEX1)                |
| PSS           | $K_S = -2$ , $T_5 = 1$ , $T_6 = 4.2$ , $T_4 = 0.75 \text{ s}$             |
| PLL           | $k_{p,\text{pll}} = 53$ , $k_{i,\text{pll}} = 1400$                       |
| Outer loops   | $\tau = 0.03 \text{ s}$ ; $k_p/k_{ip} = k_q/k_{iq} = 0.2/8$               |
| Inner loops   | $k_i/k_{ii} = 0.25/6$ ; $x_f = 0.15$ , $r_f = 0.01$                       |
| Q/V regulator | $k_{p,v}/k_{i,v} = 2/20$ ; leak $w = 0.05 \text{ rad/s}$ ; $g \in [0, 1]$ |
| Loads         | constant power                                                            |
| Core          | SGs 30, 33, 35, 37: 2096.6 MW active dispatch on 4270.7 MVA of rating     |

**Other benchmarks (same code; corroboration, not independent implementation).** Kundur two-area (policy dependence recurs; aperiodic Q/V crossings); documented IEEE-68 (power flow reproduced to  $5 \times 10^{-5}$ , 15/15 modes within  $5 \times 10^{-4}$  Hz).

## 4 Theory

### 4.1 Exact transverse quotient

Let  $R_x$  be 1 on every rotor and PLL angle and 0 elsewhere, and  $w$  the uniform frequency shift (speed +1, PLL integrator  $+\omega_B$ ).

**Theorem 1** (Transverse quotient; PROVED, classical symmetry reduction). *If device angles enter only through  $Ve^{-j\cdot}$  (rotation covariance), then  $AR_x = 0$ . If moreover  $D = 0$ ,  $P_m$  constant and the network is frequency independent, then  $Aw = \omega_B R_x$ ; hence  $C = \text{span}\{R_x, w\}$  is  $A$ -invariant with Jordan block  $\begin{bmatrix} 0 & \omega_B \\ 0 & 0 \end{bmatrix}$ , and for any orthonormal  $Z$  spanning  $C^\perp$ ,  $A_\perp = Z^\top AZ$ ,*

$$\det(sI - A) = s^2 \det(sI - A_\perp). \quad (11)$$

*The reduced field satisfies  $r(x + \phi R_x) = r(x)$ ,  $r(x + cw) = r(x) + c\omega_B R_x$ , so  $\dot{y} = Z^\top r(x^* + Zy)$  is the exact nonlinear transverse system. With  $D \neq 0$  or governors only  $C = \text{span}\{R_x\}$  remains.*

The rotation is a gauge; the drift is a physical consequence of absent primary frequency control. *Transverse stability:*  $\alpha_\perp(S, \theta) = \max \text{Re } \sigma(A_\perp) < 0$ . No eigenvalue is discarded by magnitude; a three-state numerical classifier labels unresolved points instead of guessing.

### 4.2 Minimal incompatibility hypergraph and planning

**Definition 1.** With a stable base,  $\mathcal{U}(\theta) = \{S : \alpha_\perp(S, \theta) \geq 0\}$ ;  $\mathcal{H}(\theta)$  is the family of inclusion-minimal elements of  $\mathcal{U}$ ;  $\kappa(\theta) = \min_{e \in \mathcal{H}} |e|$  ( $\infty$  if empty).

**Proposition 1** (Structure, any-order safety; PROVED, elementary).  *$\mathcal{H}$  is an antichain; a portfolio containing no hyperedge is stable;  $\mathcal{U}$  need not be upward closed. For a target  $T$ : (A)  $T$  stable; (B) some one-at-a-time chain is stable throughout; (C) every  $R \subseteq T$  stable. Then  $C \Rightarrow B \Rightarrow A$ , no converse in general, and  $T$  satisfies  $C$  iff it contains no hyperedge.*

**Theorem 2** (Boundary localization; PROVED). *Along a continuous policy path with  $g_z$  nonsingular,  $\mathcal{H}$  is constant on every interval on which no  $A_\perp(S)$  has an imaginary-axis eigenvalue; every change of  $\mathcal{H}$  is bracketed by such a crossing.*

**Supporting results (PROVED, credited).** Monotone (Metzler, order-preserving) class  $\Rightarrow A=B=C$  (Perron–Frobenius); IEEE-39 is outside it (three robust re-stabilizing pairs). Minimum-cardinality destabilization is NP-complete (sparse-PCA clique reduction; no priority claimed). Resilience complex  $K(\rho)$ : minimal non-faces  $= \mathcal{H}_{NL}(\rho)$  (Stanley–Reisner, tautological).

### 4.3 Descriptor realization, network closure, principal minors

Under matched dispatch every candidate bus carries both devices (SG weight  $1 - \delta_i$ , GFL weight  $\delta_i$ ); an absent device is a stable filler.

**Lemma 1** (Affinity).  *$J(\delta) = J_0 + \sum_i \delta_i J_i$  exactly, each  $J_i$  bus-local. The reduced  $A(\delta)$  is not affine.*

Port  $T_S(s) = g_z + g_x(sI - f_x)^{-1}f_z$  (network variables kept). With  $E$  selecting the  $2m$  voltage channels of candidate buses,  $\Delta Y_i = E_i^\top (T_{\{i\}} - T_0)E_i$  ( $2 \times 2$ ),  $D = \text{blkdiag}(\Delta Y_i)$ ,  $K = E^\top T_0^{-1}E = K_d + K_o$  (diagonal and off-diagonal  $2 \times 2$  blocks),

$$M = DK, \quad Q = (I + DK_d)^{-1}DK_o. \quad (12)$$

**Theorem 3** (Network-closure factorization; PROVED, Sylvester + multilinearity).

$$\det P_S = h_S \det T_0 \det(I + M_{SS}), \quad (13)$$

$$\det(I + M_{SS}) = \prod_{i \in S} \det(I + M_{ii}) \det(I + Q_{SS}), \quad (14)$$

$$\det(I + Q_{SS}) = \sum_{T \subseteq S} \mu_T, \quad \mu_\emptyset = 1, \quad \mu_{\{i\}} = 0, \quad \mu_T = \sum_{U: \text{blocks}(U)=T} \det Q_{UU}. \quad (15)$$

$\mu_T$  is the Boolean Möbius coefficient and is invariant under block-diagonal port-basis changes.

**Corollary 1** (Boundaries are network-closure zeros). *If no member alone and no device has an eigenvalue at  $s^*$ :  $s^* \in \sigma(A_\perp(S))$  iff  $-1 \in \sigma(M_{SS}(s^*))$  iff  $\det(I + Q_{SS}(s^*)) = 0$ . With  $K_o = 0$  no boundary exists.*

The algebraic degree of  $S \mapsto \det(I + Q_{SS})$  and  $\kappa$  are independent quantities.

## 4.4 Boundary motion (port derivative)

**Proposition 2** (Boundary sensitivity; PROVED, implicit function theorem). *If  $-1$  is a simple eigenvalue of  $M_{SS}(s^*, \theta^*)$  with right/left eigenvectors  $q, p$ ,*

$$\frac{\partial s^*}{\partial \theta} = -\frac{p^H \partial_\theta M_{SS} q}{p^H \partial_s M_{SS} q}, \quad (16)$$

*and  $\nabla_\theta \text{Re } s^*$  is the boundary normal. Equivalently, at a simple zero of  $F(H) = \det(I + Q_{HH})$ :  $\partial s^* / \partial a = -\partial_a F / \partial_s F$ , with  $\partial_a F = \text{tr}(\text{adj}(I + Q) \partial_a Q)$  valid at the singular point.*

## 4.5 Symmetry-deflated zero-frequency port

$\det T_S(0)$  vanishes structurally. Relocate  $f_x^\# = f_x - \beta U U^\top$  ( $U = R_x / \|R_x\|$ ), then deflate the Jordan partner  $x_0 = w + a R_x$ :  $f_x^{\#\#} = f_x^\# - \beta_2 x_0 x_0^\top / \|x_0\|^2$ ,  $T^{\#\#}(0) = g_z - g_x (f_x^{\#\#})^{-1} f_z$ .

**Theorem 4** (PROVED; Brauer + Schur).  $\sigma(A^{\#\#}) = \sigma(A_\perp) \cup \{-\beta, -\beta_2\}$ ;  $\det T^{\#\#}(0) = \det g_z \det A^{\#\#} / \det f_x^{\#\#}$ ; *its sign flips iff the parity of positive real transverse eigenvalues changes. It does not see complex crossings or real-pair coalescences.*

## 4.6 Finite-amplitude connected cumulants (secondary layer)

For  $F_s(S) = \det(I + Q_{SS}(s))$ ,  $F(\emptyset) = 1$ , and  $\Pi(S)$  the set partitions of  $S$ :

$$\chi_S = \sum_{\pi \in \Pi(S)} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} F(B). \quad (17)$$

All statements PROVED in `theory/FINITE_AMPLITUDE_CONNECTED_CUMULANTS.md`:

- **C1 inversion**:  $F(S) = \sum_\pi \prod_{B \in \pi} \chi_B$ , unique (exponential formula in  $\mathbb{C}[x]/(x_i^2)$ ).
- **C2**:  $F$  factorizes across  $\{S_1, S_2\}$  iff all mixed cumulants vanish;  $\chi_S = 0$  alone does not imply factorization.
- **Connectivity**: if  $F(H) = 0$  and  $F(B) \neq 0$  for all proper  $B$ , the support  $\{B \subseteq H : |B| \geq 2, \chi_B \neq 0\}$  is connected and spans  $H$ ; boundary identity  $\chi_H(s^*) = -C_H(s^*)$ ,  $C_H = F(H) - \chi_H$ .
- **C3 scalar ports**:  $\chi_S = (-1)^{|S|-1} \sum_\gamma \prod_{(i,j) \in \gamma} q_{ij}$  over spanning directed cycles for  $|S| \geq 2$  ( $\chi_{\{i\}} = 1 + q_{ii}$ ).
- **C3  $2 \times 2$  ports**: the scalar formula does not transfer;  $\chi_S$  is the sum over block-connected channel permutations;  $\chi_{ab} = -\text{tr}(Q_{ab} Q_{ba}) + \det Q_{ab} \det Q_{ba}$ . The naive holonomy sum is only the infinitesimal cumulant.
- **C4**: Boolean degree  $d_{\text{alg}}$ , connected order  $d_{\text{conn}}$  and  $\kappa$  are logically independent (ring, path, star, tuned, ring+pair constructions).
- **Connected share** at a zero:  $\nu_H = |\chi_H| / \sum_\pi |\prod \chi_B| \leq 1/2$ ; deletion displacement  $\delta_B = |\chi_B F(H \setminus B)| / |\partial_s F(H)|$ .

**IEEE-39 verdict (preregistered)**: the flagship boundary is *composite* ( $\nu = 0.069 < 0.10$ ; deleting  $\chi_{30,33,35,37}$  moves the zero by  $0.024 \text{ s}^{-1} < 0.05$ ); triple boundaries are connected at 0.10 (not at 0.20). Hence  $\kappa = 4$  is *not* an irreducible four-way interaction. Connected cumulants are a mathematically valid, diagnostically interesting, practically limited layer.

## 4.7 Second-order consistency (methodological support)

A second-order reduced response is consistent only if the algebraic-manifold (KCL) curvature  $f_z D^2 \psi$  is retained; dropping it breaks the rotation/drift symmetry and makes the frequency prediction about  $10^3$  times worse than linear (FC07 v2). Taylor consistency, not new.

# 5 Experiment catalogue

| ID      | Question / content                                                                                     | Key output (status)                                                                                                                                        |
|---------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F1      | Equation-equivalent ANDES reconciliation (R2 manual, R3 first-order AVR; static-injection replacement) | band match $1.6 \times 10^{-6}$ ; RHP 12/12; flagship +0.329 in both tools                                                                                 |
| F7A/B/C | Policy maps: $g \times k$ , $g \times t$ , $g \times h$ ; 345 229 points                               | 30/36/16 distinct $\mathcal{H}$ ; witness contraction $4 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \emptyset$ on F7B $t = 0.852$ |
| F8      | Mechanism points P1–P4, $P_{\text{fold}}$ , $P_\infty$                                                 | P4: $\mathcal{H} = \{30+33+35+37\}$ , $\alpha_\perp = +0.127$ ; $P_\infty$ : $\emptyset$                                                                   |

| ID             | Question / content                                         | Key output (status)                                                                                                                                                           |
|----------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F10            | Conventional baselines on 52 policy points                 | modal sensitivity 22/52, additive 30/52, pairwise 38/52 exact $\mathcal{H}$ ; SCR/gSCR policy-blind; GN on every sub-loop 52/52 (= our computation)                           |
| F11            | Computational scaling                                      | port core 22–119× faster <i>at a common operating point</i> only                                                                                                              |
| F12            | Kundur second benchmark                                    | policy dependence 61/61, 36/36 lines                                                                                                                                          |
| G1             | Whole-RHP composability, condensers                        | damped condenser threshold 2.48% (106.0 MVA); undamped condenser unstable                                                                                                     |
| G2             | Nonlinear phasor TDS confirmation                          | 32/32 verdicts (P4 unstable, $P_\infty$ stable, condenser 2%/3%, tongue)                                                                                                      |
| G3             | IEEE-68                                                    | documented model reproduced; preregistered map EMPTY                                                                                                                          |
| E12/E13/E39    | 512-portfolio census; conventional predictors              | MW/MVA/SCR/gSCR/MIIF AUCs (re-audited in UC02)                                                                                                                                |
| E14 N6 / FC04  | Pg-matched controls                                        | 25/25 controls keep the inter-area family stable; 8/10 failing have it unstable                                                                                               |
| E18            | Cycle holonomies                                           | falsified as explanation (Pg rerun ratio 0.72)                                                                                                                                |
| UC01–03        | Unit correction                                            | legacy “replaced_mw” is MVA; true-dispatch AUC 0.57                                                                                                                           |
| BC01–02b       | Binary certification, principal minors                     | $d^* = m$ on all benchmarks; zero-frequency port 194/194 (method development)                                                                                                 |
| FC01           | Transverse re-audit                                        | 0 label changes over 345 229 points; 1.5% unresolved                                                                                                                          |
| FC02           | Zero-frequency port holdout (Kundur)                       | 28/28 zero crossings, coalescence not flagged, 0/445 false positives                                                                                                          |
| FC03           | Governed replication (TGOV1N)                              | P4 flagship $+0.127 \rightarrow -0.0745$ ; 10 vs 26 distinct $\mathcal{H}$                                                                                                    |
| FC05/06        | Nonlinear thresholds, resilience complex                   | $\mathcal{H}_{NL} = \mathcal{H}$ inside scope 6/6; 84/96 thresholds are guard events                                                                                          |
| FC07           | Second-order curvature                                     | KCL curvature required                                                                                                                                                        |
| FC08/09        | Certificates                                               | failed as useful certificates                                                                                                                                                 |
| FC10           | Monotone class, order dependence                           | 3 robust counterexamples; 3/327 order-dependent targets                                                                                                                       |
| FC12           | Planning                                                   | any-order-safe plans; 106 vs 684–792 MVA support gap (envelope)                                                                                                               |
| FC13           | Hopf type                                                  | subcritical, $\ell_1 = +0.0095, +0.025, +0.026$                                                                                                                               |
| FC18           | Targeted port checks at 10 frozen boundaries               | identity $1.4 \times 10^{-13}$ ; truncation below $ S $ leaves 0.13–0.60; normals $4.3 \times 10^{-4}$ deg; Newton retune $2 \times 10^{-10}$ ; walk expansion non-executable |
| MC01–04        | Preregistered Monte Carlo (S1–S7 synthetic, P1–P4 IEEE-39) | all pass except P2 as implemented (post-hoc correction disclosed)                                                                                                             |
| Cross-tool A–G | Python / pandapower / ANDES                                | network, equilibrium, synchronous dynamics, branch sensitivities reproduced; custom GFL not                                                                                   |
| CC02–CC06      | Connected cumulants (toy + frozen IEEE-39)                 | flagship composite; $ \chi $ AUC 0.952 $\approx$ closure 0.944; H5, H6 refuted                                                                                                |

## 6 Key results

### 6.1 The canonical replacement-portfolio trap (frozen FC03 values)

Transverse spectral abscissa  $\alpha_\perp$  [s<sup>−1</sup>] of all 16 core subsets; frozen model versus the model with the documented TGOV1N governors.

| Subset             | P4 ( $g, k$ ) = (0.03625, 1.425) |          | P $_{\infty}$ ( $g, k$ ) = (1, 0.5) |          |
|--------------------|----------------------------------|----------|-------------------------------------|----------|
|                    | frozen                           | governed | frozen                              | governed |
| BASE               | -0.1441                          | -0.2385  | -0.1759                             | -0.2385  |
| 30                 | -0.1542                          | -0.2385  | -0.1680                             | -0.2386  |
| 33                 | -0.1696                          | -0.2387  | -0.1590                             | -0.2386  |
| 35                 | -0.1734                          | -0.2389  | -0.1662                             | -0.2387  |
| 37                 | -0.1480                          | -0.2386  | -0.1693                             | -0.2386  |
| 30+33              | -0.1821                          | -0.2074  | -0.1535                             | -0.2388  |
| 30+35              | -0.1725                          | -0.2242  | -0.1606                             | -0.2392  |
| 30+37              | -0.1511                          | -0.2213  | -0.1654                             | -0.2386  |
| 33+35              | -0.2140                          | -0.2390  | -0.1608                             | -0.2388  |
| 33+37              | -0.1768                          | -0.2389  | -0.1550                             | -0.2387  |
| 35+37              | -0.1721                          | -0.2387  | -0.1622                             | -0.2389  |
| 30+33+35           | -0.2047                          | -0.1368  | -0.1568                             | -0.2392  |
| 30+33+37           | -0.1755                          | -0.1868  | -0.1522                             | -0.2388  |
| 30+35+37           | -0.1679                          | -0.2135  | -0.1598                             | -0.2391  |
| 33+35+37           | -0.2030                          | -0.2387  | -0.1574                             | -0.2390  |
| <b>30+33+35+37</b> | <b>+0.1270</b>                   | -0.0745  | -0.1554                             | -0.2390  |

At P4 every proper subset is stable and the complete portfolio is unstable:  $\kappa = 4$ ,  $\mathcal{H} = \{\{30, 33, 35, 37\}\}$  (critical mode 0.62 Hz, inter-area). The same physical portfolio, same network, same injections and ratings is stable at P $_{\infty}$  ( $\mathcal{H} = \emptyset$ ). With documented governors the P4 portfolio is stable. Source: `outputs/ias2026/final_math_nonlinearity_validation_20260910T231539/FC03_governed_replication/FC03_subsets.csv.gz`. Nonlinear phasor TDS confirms P4 unstable (envelope  $+0.129 \text{ s}^{-1}$  vs  $+0.127$  predicted) and P $_{\infty}$  stable (G2).

**Lower-order screens at P4 (frozen F10).** First-order modal sensitivity, additive extrapolation from single studies, and pairwise extrapolation from single+pair studies all predict  $\mathcal{H} = \emptyset$  (“safe”) at P4, where the truth is the complete four-unit portfolio. Over 52 policy points they recover the exact  $\mathcal{H}$  in 22, 30 and 38 cases respectively. SCR/gSCR are identical at P4 and P $_{\infty}$  by construction.

## 6.2 Policy dependence (C1)

- 30/36/16 distinct hypergraphs on F7A/B/C; up to 15 hypergraphs behind one value of  $\kappa$ .
- F7B line  $t = 0.852$ : witness  $4 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \emptyset$  as  $g$  increases.
- P4 line ( $k = 1.425$ ,  $t = 1.5$ ): the complete portfolio is unstable for  $g < g^* = 0.20768$  and stable above;  $\kappa = 4$  for  $g \in [0.016, g^*)$ , while below  $g \approx 0.016$  the triple 30+33+35 is the witness (frozen FC03 path). This is a converter-only counterfactual: only the Q/V gain changes.
- Governed model: policy dependence persists (10 distinct  $\mathcal{H}$ ,  $\kappa = 2$ –4 at 74/399 plane points) but the P4 coalition is stabilized.
- Kundur: 61/61 and 36/36 lines change  $\mathcal{H}$ . IEEE-68: preregistered map empty.

## 6.3 Network-closure anatomy (C2, FC18)

| Event                       | $S$ changing | $g^*$  | Hz    | local factor | Boolean trunc. at $ S  - 1$ |
|-----------------------------|--------------|--------|-------|--------------|-----------------------------|
| E1 $\kappa 4 \rightarrow 3$ | 30+33+35     | 0.0102 | 0.628 | 0.315        | 0.336                       |
| E2                          | 30+33+37     | 0.0188 | 0.633 | 0.198        | 0.396                       |
| E3                          | 33+35+37     | 0.0609 | 0.671 | 0.080        | 0.385                       |
| E4 $\kappa 3 \rightarrow 2$ | 30+33        | 0.1034 | 0.656 | 0.081        | 1                           |
| E5                          | 33+35+37     | 0.1870 | 0.695 | 0.135        | 0.296                       |
| E6 $\kappa 2 \rightarrow 3$ | 30+33        | 0.2491 | 0.672 | 0.104        | 1                           |
| E7                          | 30+33+35     | 0.2778 | 0.706 | 0.130        | 0.372                       |
| E8 $\kappa 3 \rightarrow 4$ | 30+33+37     | 0.2931 | 0.685 | 0.132        | 0.293                       |
| E9 $\rightarrow \emptyset$  | 30+33+35+37  | 0.2973 | 0.718 | 0.134        | 0.132                       |
| FLAG (P4 line)              | 30+33+35+37  | 0.2077 | 0.706 | 0.089        | 0.161                       |

Identity  $\det T_S / \det T_0 = \det(I + M_{SS})$  to  $1.4 \times 10^{-13}$  (160 checks); descriptor affinity to  $4.5 \times 10^{-16}$ ; effective rank of  $Q$  stays 7–8 while  $\kappa$  changes (negative control). Caveat from CC03: at the flagship the full-order Boolean term is composite in cumulant terms and its eigenvalue weight is only  $0.024 \text{ s}^{-1}$ .

## 6.4 Boundary motion and remediation (C3)

- Port normals vs direct finite differences: angle  $\leq 4.3 \times 10^{-4} \text{ deg}$ , magnitude  $\leq 4.3 \times 10^{-6}$ .

- Q/V retuning of P4: one linear step predicts  $\Delta g = 0.052$  (30% of the required 0.171; the slope drifts from  $-2.45$  to  $-0.46$ ); port-derivative Newton reaches  $g^* = 0.20768$  to  $2 \times 10^{-10}$  in 5 steps.
- Damped condenser: stability restored above 2.48% of the retired rating (106.0 MVA); TDS 2.47%.
- Documented governors: P4 flagship stable without support.
- Line reinforcement (cross-tool E1 / F2c): port line sensitivities reproduce the DAE (12/12 signs, Spearman 1.00) and ANDES in the equation-equivalent configuration; the GFL line ranking does not transfer across converter models (7/12 signs).

## 6.5 Planning

Any-order-safe optimum: P4 core 30+33+35 (1775.1 MW / 3300.5 MVA);  $P_\infty$  core full set (2096.6 MW / 4270.7 MVA); census 31+32+34+35+37+38 (3652.3 MW / 6499.9 MVA). Order dependence: 3 of 327 stable census targets (unit 34 converted last passes through an unstable state). Support gap: 106 MVA (small-signal) vs 684–792 MVA (keeping a 200 MW disturbance inside the declared converter voltage envelope; an envelope requirement, not stability).

## 6.6 Nonlinear scope check

Inside declared guards,  $\mathcal{H}_{NL} = \mathcal{H}$  and  $\kappa_{NL} = \kappa$  in 6/6 (point, family) cases; no transversely stable portfolio fails in scope; 84/96 thresholds are guard activations. Boundaries are subcritical Hopf. TDS verdicts: 32/32 (G2) and 80/80 (MC P4, rate vs  $\alpha_\perp r = 0.969$ ).

## 6.7 Monte Carlo (preregistered)

S1 (1000 draws) transverse identity  $\leq 5.5 \times 10^{-9}$ , magnitude cutoff wrong in 235/235 planted positives; S2 (2000) 0 violations; S3 (300) 1044 crossings bracketed; S4 Metzler 0 violations; S5 500/500; S6 identities  $\leq 3.4 \times 10^{-15}$ ; S7 817/817. IEEE-39: P1 300/300; P3 identities  $\leq 8.1 \times 10^{-13}$ ; P4 80/80; P2 failed as implemented (post-hoc re-bisection  $\leq 2 \times 10^{-9}$ , disclosed, not counted).

## 6.8 Cross-tool validation (commit 0a4e18c2)

- Network/equilibrium: ANDES 2.0.0 live Ybus  $1.1 \times 10^{-13}$ ; voltages  $1.7 \times 10^{-7}$  (entirely ANDES’s  $+10^{-8}$  r/x regularization); pandapower 3.4.0  $3 \times 10^{-14}$  *after* a tap-convention translation fix (as supplied: FAIL).
- Equation-equivalent synchronous dynamics (R2/R3, 12 subsets):  $\alpha$  error  $1.5 \times 10^{-6}$ , band nearest-mode  $1.6 \times 10^{-6}$ , RHP counts agree.
- Equation-equivalent branch sensitivities (12 frozen lines): DAE vs ANDES 12/12 signs, Spearman 1.000; port vs ANDES 11/12 (the only mismatch on a near-zero line).
- Custom GFL: **not** independently implemented.

## 6.9 Connected-cumulant results (commit cdfcefcf)

Ten boundaries: reconstruction  $\leq 10^{-16}$ ; connectivity 10/10; triples connected at 0.10; flagship and E9 composite. Diagnostic:  $|\chi|$  AUC 0.952 (failing vs Pg-matched stable), closure distance 0.944, best older non-definitional diagnostic 0.812 ( $|\mu|$ ), simple-cycle holonomy 0.084 (anti-predictive). “Dominant cluster tracks the witness” refuted (34%); “best stabilizing line suppresses  $|\chi|$ ” refuted (Spearman  $-0.34$ ).

## 7 Claim ledger (condensed)

| ID  | Claim                                                                                   | Status                              |
|-----|-----------------------------------------------------------------------------------------|-------------------------------------|
| T01 | Exact transverse quotient; exact nonlinear transverse system                            | PROVED                              |
| T02 | $\mathcal{H}$ antichain; hyperedge-free $\Rightarrow$ stable; changes only at crossings | PROVED                              |
| T03 | Any-order safety iff hyperedge-free; $C \Rightarrow B \Rightarrow A$                    | PROVED                              |
| T04 | Metzler monotone class $\Rightarrow A=B=C$ (IEEE-39 outside it)                         | PROVED                              |
| T05 | NP-completeness of minimum destabilizing cardinality (credited)                         | PROVED                              |
| T06 | Resilience complex organization                                                         | PROVED (tautological)               |
| T07 | Principal-minor / Möbius structure and invariants                                       | PROVED                              |
| T08 | Symmetry-deflated zero-frequency port                                                   | PROVED; Kundur holdout 28/28, 0/445 |
| T09 | Network-closure anatomy at 10 frozen boundaries                                         | BENCHMARK-SPECIFIC                  |

| ID                 | Claim                                                                                                                                                                               | Status                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| T10                | Port derivative = DAE eigenvalue sensitivity                                                                                                                                        | NUMERICALLY<br>VALIDATED;<br>independently reproduced<br>(equation-equivalent) |
| C01–<br>C06<br>C07 | Connected-cumulant theorems (inversion, factorization, connectivity,<br>scalar cycle formula, block expansion, order independence)<br>Flagship boundary composite ( $\nu = 0.069$ ) | PROVED<br>BENCHMARK-<br>SPECIFIC                                               |
| C08                | Triple boundaries connected at 0.10                                                                                                                                                 | BENCHMARK-<br>SPECIFIC<br>(threshold-sensitive)                                |
| C09                | $ \chi $ discriminates (AUC 0.952), equals closure distance                                                                                                                         | BENCHMARK-<br>SPECIFIC                                                         |
| C10                | Dominant connected support tracks the witness                                                                                                                                       | REFUTED                                                                        |
| C11                | Most stabilizing intervention suppresses $ \chi $                                                                                                                                   | REFUTED                                                                        |
| B01                | Policy-dependent minimal incompatibility                                                                                                                                            | CONDITIONAL (custom<br>GFL, no governor)                                       |
| B02                | Flagship $\kappa = 4$ at P4, $\emptyset$ at $P_\infty$                                                                                                                              | CONDITIONAL<br>(removed by governors)                                          |
| B03                | Governed replication                                                                                                                                                                | BENCHMARK-<br>SPECIFIC                                                         |
| B04–<br>B06        | Inter-area mechanism; condenser 106 MVA; 3/327 order-dependent                                                                                                                      | BENCHMARK-<br>SPECIFIC                                                         |
| B07–<br>B10        | Nonlinear = spectral in scope; subcritical Hopf; TDS 32/32; KCL<br>curvature                                                                                                        | NUMERICALLY<br>VALIDATED                                                       |
| I01                | Network/equilibrium implementation-independent                                                                                                                                      | CONDITIONAL<br>(pandapower fix)                                                |
| I02,<br>I03        | Equation-equivalent synchronous dynamics; branch sensitivities                                                                                                                      | NUMERICALLY<br>VALIDATED (ANDES)                                               |
| I04                | Custom-GFL portfolio results reproduced in a second simulator                                                                                                                       | NOT YET TESTED                                                                 |
| N01–<br>N09        | Simple cycles; static quantities alone; universal order 4; robustness to<br>governors; rank; paths; certificates; low-degree representation; transferable<br>GFL line ranking       | REFUTED                                                                        |

## 8 Negative results (preserved)

- Simple-cycle causality failed (E18/UC02); the single-cycle holonomy score is anti-predictive (AUC 0.084).
- Static topology or size alone fails: true-dispatch AUC 0.57; SCR/gSCR cannot see the policy.
- Order four is model- and policy-dependent: witness contraction, governors, IEEE-68 empty, composite flagship.
- Governors change the witness: the P4 coalition is stable with TGOV1N.
- ANDES does not implement the custom GFL.
- Rank does not explain  $\kappa$ ; the walk/path expansion is non-executable; certificates failed; no exact low-degree Boolean representation.
- Connected cumulants: not the mechanism of the flagship; do not track the witness; not a design target.
- P2 Monte Carlo rule failed as implemented; the pandapower translation failed as supplied.

## 9 What is missing

### 9.1 Pending final validation campaign (specified, not yet run)

The last instruction defines one final campaign, *before* any new “weak-node” theory. Status: Phase 0 reading done; preregistration **not yet written**; nothing computed.

1. **Preregistration** (docs/20260911\_POST\_CUMULANT\_VALIDATION\_PREREG.md) committed before any computation.
2. **Industrial decision case** on the frozen P4 portfolio: D1 admissible? D2 minimum blocking coalition? D3 predictable from MW/MVA, static strength, single-device modal data, pairwise data, topology? D4 intrinsic or policy-dependent? D5 which targeted intervention restores stability? D6 robust to uncertainty? Terminology: “budget-limited subset screening”, not “N-k”.
3. **Fair baselines** B0 oracle, B1 MW/MVA/penetration, B2 SCR/gSCR/MIIF, B3 first-order modal sensitivity, B4 additive, B5 pairwise (and third order), B6 participation (localization only), B7 static network metrics,

B8 closure distance (operator-only and oracle-assisted versions), B9  $|\chi|$  (secondary). Old counts (22/52, 30/52, 38/52, AUC 0.57) must be re-verified from frozen files.

4. **Preregistered metrics:** Task A ranking (AUC, AP, Spearman), Task B minimum-set recovery, Task C margin (sign, MAE/RMSE), Task D policy transfer P4 vs  $P_\infty$ , Task E intervention ranking; N/A where a method does not answer the question.
5. **Policy counterfactual:** what stays constant (network, locations, injections, ratings, MW, SCR/gSCR) and what changes ( $\alpha_\perp$ ,  $\mathcal{H}$ ,  $\kappa$ , frequency). Note that  $P4 \rightarrow P_\infty$  changes both  $g$  and  $k$ ; a converter-only counterfactual exists on the frozen P4 line ( $k = 1.425$ ,  $g \geq 0.2256$  gives  $\emptyset$ ).
6. **Why lower orders fail:** hierarchy local  $\rightarrow$  additive  $\rightarrow$  pairwise  $\rightarrow$  third order  $\rightarrow$  exact, in both  $\alpha_\perp$  space and characteristic-function space; distinguish  $\kappa = 4$  from the (composite) connected order.
7. **Targeted remediation:** Q/V gain, excitation gain  $k$  (frozen F7A line), condenser, governors, frozen lines; restored?  $\Delta\alpha_\perp$ ? port prediction? Newton boundary? TDS corroboration?
8. **Parameter robustness** (optional, high value): R1 existence robustness vs R2 exact-witness robustness, with the uncertainty set committed before running.
9. **Independent custom-GFL reproduction** (highest remaining external validation): equation-equivalent implementation of the 10+1-state GFL, the two-axis machine, first-order AVR and power-input PSS in a *separate* ANDES copy and environment (the frozen `tx3-andes` must not change); preregistered tolerances; 16 subsets at P4 and  $P_\infty$  at minimum. Feasibility: ANDES substitutes static generators through the dynamic device's `gen` link; IEEEEST can express washout+lag with power input if configured exactly.
10. **Reviewer decision table**, one four-panel decision figure (no cumulant hero panel), final validation ledger with columns for proof, synthetic, IEEE-39, nonlinear, independent tool, parameter robustness, baseline comparison, status and exact manuscript wording.

## 9.2 Open manuscript decisions (author)

1. **pandapower fix:** accept or reject the tap-convention translation correction (three LV-from transformers 35, 37, 38 reoriented with tap  $1/t$  and  $z t^2$ ). If rejected, cite only the Python–ANDES static agreement.
2. **Limitations sentence** (`main.tex:959`) is stale (“within 4.1%”); replace with the equation-equivalent cross-tool wording.
3. **Clarify the C2 wording:** the full-order Boolean closure term is exact but is not an irreducible four-way interaction (flagship composite in cumulant terms).
4. Affiliation/e-mail, funding, repository/DOI, page policy (11 pp).

## 9.3 Older gaps still open (not blocking)

Machine-parameter sensitivity map (E37: full pattern in 418 of 1000 draws); dispatch-dependence map; adaptive per-point retune; extended-precision spot check; a second benchmark with an order-4 core (IEEE-68 map empty); limiter-equipped, governed and mixed GFM models (new, separately versioned).

## 9.4 Future study (only after the campaign)

A separate preregistered study of weak nodes, weak corridors and intervention locations.

# 10 Where everything is

| Item                       | Location / commit                                                                                |
|----------------------------|--------------------------------------------------------------------------------------------------|
| Scientific evidence freeze | tag IAS2026_FINAL_SCIENTIFIC_EVIDENCE_FREEZE = b9f274e2                                          |
| Post-freeze addenda        | f775db89 (FC18), 7a772808/f2946257 (MC prereg), cfe9fe4b (MC + manuscript), 81ddb963 (hardening) |
| Cross-tool validation      | 0a4e18c2; validation_runs/cross_tool_final/                                                      |
| Connected cumulants        | de09db3b (theory, prereg), cdfcefcf (results, ledger)                                            |
| Model code                 | src/ibr_cycles/models/ (ieee39_case.py, ieee39_devices.py, ieee39_network.py)                    |
| Transverse / port code     | src/ibr_cycles/certification/, src/ibr_cycles/models/port_admittance.py                          |
| Cumulant code              | src/ibr_cycles/cycles/connected.py (finite), cumulants.py (infinitesimal)                        |
| Frozen FC outputs          | outputs/ias2026/final_math_nonlinear_validation_20260910 T231539/                                |
| Policy maps                | results/F7/ (F7A/B/C points, labels, lines)                                                      |

| Item             | Location / commit                                                                                                                                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baselines        | <code>results/F10/</code> , <code>results/UC/UC02/</code> , <code>results/tables/E13_*</code>                                                                                        |
| Monte Carlo      | <code>results/paper_mc/</code> , <code>experiments/paper_tx4/</code>                                                                                                                 |
| Environments     | <code>.venv/tx3-analysis</code> (frozen), <code>.venv/tx3-andes</code> (frozen, ANDES 2.0.0 from <code>vendor/andes</code> ), <code>.venv/xtool-pandapower</code> (pandapower 3.4.0) |
| Numerical gotcha | set <code>OPENBLAS/OMP/MKL_NUM_THREADS=1</code> in the shell before multiprocessing runs                                                                                             |